崑山科技大學日間部電子工程系四技綠能科技技優領航專班課程科目表

Kun Shan University 4-Year College Curricula

課程名稱(講授時數—實習時數—學分數)

Course Name (Lecture Hours – Practice Hours – Credits)

112.11.02 系課程委員會通過
112.11.15 院課程委員會通過
113.01.24 校課程委員會通過

113 學年度入學適用 Apply to the freshmen enrolled in 113

第一學年 First Academic Year		第二學年 Second Academic Year		第三學年 Third Academic Year		第四學年 Fourth Academic Year		
第一學期 First Semester	第二學期 Second Semeste	第一學期 First Semester	第二學期 Second Semeste	第一學期 First Semester	第二學期 Second Semeste	第一學期 First Semester	第二學期 Second Semeste	
共同必修科目(合計 29 學分) University-wide Required Courses (29 Credits)								
校訂共同科目(合計 2 學分) General Courses (2 Credits)								
	綠色生活實踐教育(一) Green Living Practice (I) (0-1-1)	綠色生活實踐教育(二) Green Living Practice (II) (0-1-1)	服務學習與公民教育 Public Service and Civil Education (0-1-0)					
核心通識科目(合計 17 學分) Common Core Courses (17Credits)								
體育 (一) Physical Education (I) (2-0-2) 中文閱讀與表達(一) Chinese Reading and Expression(I) (2-0-2) 英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2)	體育 (二) Physical Education (I) (2-0-2) 中文閱讀與表達(二) Chinese Reading and Expression(II) (2-0-2) 英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2)			外語選讀 Foreign Languages ** (民創 2-0-2) 生涯發展與職業倫理 Career Development and Work Ethics (工民商 2-0-2)	外語選讀 Foreign Languages ** (工商 2-0-2) 生涯發展與職業倫理 Career Development and Work Ethics (創 2-0-2)			英文能力檢定* English Proficiency Examination* (1-0-1)
分類通識科目(合計 10 學分) Liberal Education Courses (10 Credits)								
分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)	分類通識 Liberal Education (2-0-2)				

註：Note:

*「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」辦理。

'English Proficiency Examination' is implemented according to 'Kun Shan University Guidelines for English Proficiency Requirement.'

**「外語選讀」包括「第二外語」及「職場英文」為選項式必修科目，課程擇一門修習即可。課程開設於三年級，然「第二外語」課程為二年級優先選課。

'Foreign Languages' comprises 'Foreign Languages (other than English)' and 'Workplace English', from which students choose one course. While the courses are offered in the third academic year, 'Foreign Languages (other than English)' is given preference to sophomores.

專業必修科目(合計 31 學分) Required subjects for majors (Total：31 Credits)

工程技術概論 Introduction to Engineering Technology (2-0-2)	程式設計 Program Design (1-2-3) 數位邏輯設計與實習 Digital Logic Design and Practice (2-1-3)	電子學與實習(一) Electronics and Practice(I) (1-2-3) 基本電子學實習 Electronics Practice (0-2-2) 科技英文 Scientific and Technological English (2-0-2) 微處理機基礎 Basis of Microprocessor (1-2-3)	電子學與實習(二) Electronics and Practice(II) (1-2-3)	實務專題(一) Practice and Study on Selected Topic (I) (0-1-1)	實務專題(二) Practice and Study on Selected Topic (II) (0-1-1) 智慧化科技應用實務 Practice on Intelligent Technology Application (3-0-3) 學術倫理* Academic Ethics (1-0-1)	暑期校外實習*** Summer Internship (0-＊-4)		
2-0-2	3-3-6	4-6-10	1-2-3	0-1-1	4-1-5	0-＊-4		
註*：「學術倫理」之實施與學分取得依課務組網頁之「學術倫理教育實施要點」辦理。The implementation of "Academic Ethics" and the acquisition of credits are handled in accordance with the "Implementation Points of Academic Ethics Education" on the website of the Academic Affairs Group.								
專業選修科目，依每學期實際情況開課 (選修學分 68 學分) Major elective subjects are taught according to the actual conditions of each semester (68 credits for electives)								

電子工程基礎(一) Practice Project－Fundamentals of Electronic Engineering(I) (0-3-3) 電路板佈局設計與實習 Print Circuid Board Layout Design and Practice (2-1-3) 微積分(一) Calculus(I) (2-0-2) 普通物理與實驗(一) General Physics and Experiment(I) (1-2-3)	電子工程基礎(二) Practice Project－Fundamentals of Electronic Engineering(II) (0-3-3) 互動多媒體程式設計 Interactive Multimedia Programming (3-0-3) 人工智慧概論 Introduction to Artificial Intelligence (3-0-3) 微積分(二) Calculus(II) (2-0-2) 普通物理與實驗(二) General Physics and Experiment(II) (1-2-3) 機器人工程師實務 Robot Engineer Practice (3-0-3)	電子工程進階(一) Advanced Level I in Electronic Engineering (0-3-3) 電路學 Electric Circuits (3-0-3) 工程數學 Engineering Mathmatics (3-0-3) 電動車概論 Introduction to Electric Vehicle (3-0-3)	電子工程進階(二) Advanced Level II in Electronic Engineering (0-3-3) 微算機系統設計實務 Practic on Micrcomputer System Design (2-1-3) 儀表控制軟體與模擬 Instrument Controlling Software and Simulation (2-1-3) 數位系統設計與實習 Digital System Design and Practice (2-1-3) 電動車機電整合概論 Introduction to Mechatronics of Electric Vehicles (3-0-3) AIoT 基礎與實習 AIoT Fundamental and Practice (2-1-3)	電子工程應用 Electronic Engineering Applications (0-3-3) 可程式控制器基礎及實習 Programmable Logic Controller Foundation and Practice (2-1-3) 智慧感測與雲端控制應用 Intelligent Sensing and Cloud Control Applications (3-0-3) 感測器原理與應用實務 AI Sensor and It's Application (1-2-3) 機器人智慧化實務與應用 Practice and Application of Robot Intelligence (3-0-3) 基礎量測及實習 Fundamental Measurement and Practice (2-1-3) 電動車輛工程 Electric Vehicle Engineering (2-1-3)	可程式控制器應用及實習 Application and Practice of PLC (2-1-3) 智慧機器人控制系統 Intelligent Robot Control System (3-0-3) 專業證照實習 Practical of Professional Certificate (0-1-1) 機電整合實務 Mechatronics Practice (1-2-3) 智慧製造 Intelligent Manufacturing (3-0-3) 智慧車用電子實務 Practice of intelligent Automotive Electronics (1-2-3) 車聯網原理與實作 Principles of Vehicle Intranet and the Laboratory (1-2-3) 車輛電機機械 Vehicle motor machinery (2-1-3)	人機介面實務 Practice on Human-Machine Interface (3-0-3) 學期校外實習(一) Semester Intership (I) (0-*-9) 產線智慧化實務(一) Practice on Intelligent Production Line (I) (1-2-3) 智慧生產管理 Intelligent Production Management (3-0-3) 人機介面實務 Practice on Human-Machine Interface (3-0-3) 智慧車輛控制系統 Intelligent Vehicle Control Systems (2-1-3)	學期校外實習(二) Semester Intership (II) (0-*-9) 產線智慧化實務(二) Practice on Intelligent Production Line (II) (1-2-3) 電動車檢診實務 Electric vehicle inspection practice (2-1-3)	
技優生推薦選修之三創與跨領域課程 Recommended electives for gifted students: three innovative and cross-field courses								
※三創選修課程(須修至少 18 學分) Three elective courses (required to take at least 18 credits)								

創意程式設計-C-E creative programming -C-E (2-0-2) 創意媒體與生活 Creative media and life (1-0-1)	創意故事 creative story (2-0-2) 創意管理 creative management (3-0-3)	創意電子控制工具使用入門-E Getting Started with Creative Electronic Control Tools-E (2-0-2) 創意媒體英文表達-院核心 Creative Media English Expression-House Core (2-0-2) 創意文案撰寫 Creative Copywriting (1-2-3)	創意行銷 Creative Marketing (2-0-2) 創意開發 Creative Development (2-0-2) 電腦輔助創新設計 Computer-Aided Innovative Design (1-2-3)	綠色科技與創新-C Green Technology and Innovation-C (2-0-2) 創意思考與方法 Creative Thinking and Methods (2-0-2) 廣告創意與文案 Advertising Creativity and Copywriting (1-1-2)	創新事業導論-B Introduction to Innovation Business-B (2-0-2) 創新智慧生活科技 Innovative Smart Life Technology (2-1-3)	創意思考訓練 Creative Thinking Training (2-0-2) 創意與專利 Creativity and Patent (3-0-3) 創意與創業 Creativity and Entrepreneurship (3-0-3)	就業與創業技能輔導 Employment and Entrepreneurship Skills Counseling (3-0-3) 環保與創意 Environmental Protection and Creativity (3-0-3) 創業與經營實務 Entrepreneurship and Business Practice (2-1-3)	
※跨領域選修課程(須修至少 18 學分) Interdisciplinary elective courses (at least 18 credits required)								
綠能科技與實作 Green Energy Technology and Implementation (2-1-3) 太陽能工程概論 Introduction to Solar Engineering (3-0-3) 燃料電池應技術 Fuel Cell Technology (3-0-3) 電子商務 E-Commerce (3-0-3)	工程材料學 Engineering Materials (3-0-3) 工程技術概論 Introduction to Engineering Technology (2-0-2) 遊戲企劃 Game Planning（3-0-3）	智慧車輛控制系統 Intelligent Vehicle Control System (3-0-3) 智慧製造技術 Smart Manufacturing Technology (3-0-3) 多媒體製作 Multimedia Production (3-0-3) 遊戲行為分析 Game Behavior Analysis (2-0-2)	物聯網與大數據於智慧製造應用 Internet of Things and Big Data in Smart Manufacturing Applications (3-0-3) 工業 4.0 特色技術 Industry 4.0 Featured Technologies (3-0-3) 智慧商務應用 Smart Business Applications (3-0-3) 智慧商圈管理 Smart Business District Management (3-0-3)	新能源與人文社會社會發展 New Energy and Humanistic and Social Development (2-0-2) 綠色能源概論 Introduction to Green Energy (3-0-3) 材料科學導論 Introduction to Materials Science (3-0-3) 綠建築與智慧建築 Green Building and Smart Building (3-0-3) 行動遊戲製作 Mobile Game Production (2-1-3)	儲能系統概論 Introduction to Energy Storage Systems (3-0-3) 綠色能源與產業商機 Green Energy and Industrial Business Opportunities (3-0-3) 企業節能減碳概論 Introduction to Enterprise Energy Saving and Carbon Reduction (3-0-3) 環境規劃與管理 Environmental Planning and Management (3-0-3) 能源應用 Energy Applications (3-0-3) 遊戲行銷營運實務 Game Marketing Operation Practice (0-2-2)	環工生質能源 Environmental Biomass Energy (3-0-3) 溫室氣體管理 Greenhouse gas Management (3-0-3) 光電材料 Optoelectronic Materials (3-0-3) 節能材料概論 Introduction to Energy-Saving Materials (3-0-3) 太陽光電能發電系統 Solar Photoelectric Power Generation System (3-0-3)	物聯網於綠能之應用 Application of Internet of Things in Green Energy (3-0-3) 智慧能源與物聯網 Smart Energy and Internet of Things (3-0-3) 物聯網與綠能概論 Introduction to Internet of Things and Green Energy (2-0-2) 物聯網與綠能應用概論 Introduction to Internet of Things and Green Energy Applications (3-0-3) 物聯網與綠能應用實務 Internet of Things and Green Energy Application Practice (3-0-3)	

註： 校外實習課程依電子系之「校外實習課程實施辦法」辦理 Off-campus internship courses are handled in accordance with the "Implementation Measures for Off-campus Internship Courses" of the Department of Electronic Engineering
最低畢業學分：128 學分 The minimum total credits for graduation is 128 credits
備註： 1. 本系畢業學分為 128 學分，必修 60 學分(包含共同科目、系專業必修科目)、選修 68 學分(系專業選修科目、三創選修課程、跨領域選修課程及學生自行選修經系/學位學程認可之專業課程)。The minimum graduation credits of this department are 128 credits: 56 credits of compulsory courses and 72 credits of elective courses 2. 本技優專班課程，規劃 18 學分以上的三創課程及跨領域等選修課程，協助技優生運用優秀的技術能力及創造力，呈現多元的創作能量，體現技職教育的特色。This special class for technical students plans more than 18 credits of three innovative courses and cross-field elective courses to help technical students use their excellent technical abilities and creativity to present diverse creative energy and reflect the characteristics of technical and vocational education. 3. 本系學生需至少修讀完成一跨院系學分學程，取得跨院系學分學程證書使得畢業。Students in this department need to complete at least one cross-department credit course and obtain a cross-department credit course certificate to graduate.。 4. 本系學生需至少修讀完成一數位科技微學程，取得數位科技微學程證書使得畢業。Students in this department need to complete at least one digital technology micro-course and obtain a digital technology micro-course certificate to graduate.